

Requirements Documentation

[GitHub]: https://github.com/DanielJ0131/interactive-house/tree/main/SG3_Devices/SG3_Artifacts

Subgroup 3: Devices

Interactive House Project (Device Subgroup).....

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R1: Hardware Initialization.....

R2: Real-time Environmental Sensing.....

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R7: Command Buffering.....

R8: State-Based Execution.....

R9: Authorization State Awareness.....

Interactive House Project (Device Subgroup)

Revision History

Date	Version	Description	Author
06/Feb/2026	1.0	Initial skeleton and device list.	Ryad Hazin
25/Feb/2026	1.1	Updated to reflect Keystudio Arduino hardware kit. removed simulation/GUI requirements.	Ryad Hazin
27/Feb/2026	1.2	Added R6 Serial communication requirement.	Patrick Nordahl
15/Feb/2026	1.3	Added "R7: Command Buffering" and "R8: State-Based Execution"	Tony Nguyen, Patrick Nordahl
21/April/2026	1.4	Updated R7/R8 status to Fulfilled; Added R9 (Authorization),	Tony Nguyen
23/April/2026	1.5	R3: Automated Actuator Response (UPDATED)	Mustafa Al-Bayati, Ryad Hazin, Tony Nguyen, Sergej Macut
08/May/2026	1.6	Updated file to contain Table of content.	Mustafa Al-Bayati, Ryad Hazin, Tony Nguyen,

			Sergej Macut
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Requirements List

Requirement Name	Priority	Progress
R1. Hardware Initialization	Essential	100%
R2. Real-time Environmental Sensing	Essential	100%
R3. Automated Actuator Response	Essential	100%
R4. Local Visual Feedback (LCD)	Essential	100%
R5. Manual Control (Buttons)	Desirable	100%
R6. Gateway1 Communication (Serial output)	Essential	100%
R7: Command Buffering	Essential	100%
R8: State-Based Execution	Essential	100%
R9: Authorization Awareness	Essential	0%

Requirements Descriptions

R1: Hardware Initialization

The system must initialize all digital and analog pins connected to the Keyestudio sensors and actuators (LEDs, Fan, Servo, Buzzer) upon power up.

R2: Real-time Environmental Sensing

The Arduino must continuously monitor analog and digital inputs from the hardware kit, specifically the MQ-2 Gas Sensor, PIR Motion Sensor, and Steam Sensor.

R3: Automated Actuator Response

The system shall implement autonomous behavior, where environmental sensor data triggers automatic actuator responses without requiring user interaction.

Examples include:

If gas level exceeds safety threshold → activate buzzer and fan

If rain is detected → close window + door

This ensures that the system can react in real-time to environmental conditions, even without communication with the Gateway or Server.

R4: Local Visual Feedback (LCD)

Current system data (Light ON, Fan ON, Buzzer!) must be displayed on the LCD Display.

R5: Manual Control (Buttons)

The system should allow manual override of physical components (like the yellow LED or Servo controlled door) using the physical Push Button modules included in the kit.

*Note: It is acknowledged that these **traditional physical interfaces are rigid and potentially inaccessible for individuals with limited mobility**, necessitating the inclusion of remote control through the middleman application.*

R6: Gateway Communication (Serial Output)

Arduino shall be able to transmit sensor values and current device state via Serial communication in a structured format (e.g., JSON or key=value pairs), enabling an external gateway (e.g., laptop or Raspberry Pi) to read and interpret the data.

R7: Command Buffering

Arduino firmware shall implement a non-blocking serial read buffer to ensure that if multiple commands are received in rapid succession from the Gateway, they are executed sequentially without crashing the system

Note: Requirement is primarily handled by the Subgroup 1 Gateway, which serializes incoming commands before they reach the Arduino. The local Arduino buffer now serves as a secondary safety net.

R8: State-Based Execution

Arduino firmware shall check the current hardware state before executing a command to prevent redundant operations and ensure "idempotency" (performing the same action multiple times has the same result as performing it once).

Note: Idempotency is now enforced by the Middleware, which tracks the global state of the house before transmitting a change request to the hardware.

R9: Authorization State Awareness

The system must receive and maintain a real-time "Armed/Locked and away" or "Disarmed/Unlocked and home" state from the central database. This state determines if motion detection triggers a standard convenience light or activates the **Intruder Logic** (flashing LEDs, buzzer alarm, and "DIALING 112" LCD message).